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SUPERSEDING
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MILITARY SPECIFICATION

X-RAY APPARATUS, DENTAL, WALL MOUNTED, 15 MA, 90 PKV, 100-130 VOLT, 50-60 Hz, A.C.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a wall-mounted dental X-Ray apparatus, consisting of a control unit and a separate tubearm-tubehead assembly, in which the control unit is capable of operating at least three remotely located, wall-mounted, tubearm-tubehead assemblies.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

GG-X-635	X-Ray apparatus, Radiographic and Fluoroscopic.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.
PPP-B-1055	Barrier Material, Waterproofed Flexible.

Military

MIL-M-7298	Manual, Technical, Commercial Equipment

STANDARDS

Federal

FED-STD-595	Colors
FED-STD-83	X-Ray Tube Focal Spot, Method of Measurement

FSC 6525

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Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNDERWRITERS' LABORATORIES, INC.

UL 187 Standards for Safety-X-Ray Equipment.

(Application for copies should be addressed to the Underwriters' Laboratories, Inc., 1285 Walt Whitman Road, Melville, L.I., New York 11746, 207 East Ohio Street, Chicago, Illinois 60611, 1655 Scott Boulevard, Santa Clara, California 95050, 333 Pfingsten Rd., Northbrook, Ill. 60062.)

National Council on Radiation Protection and Measurements:

NCRP Report No. 35 Dental X-Ray protection.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D. C. 20402.)

American National Standards Institute: (ANSI)

B1.1 - Unified Screws Threads.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, New York 10018.)

International Electrical Commission (IEC)

Publication 336 Measurement of focal spots

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, New York 10018.)

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U.S. Dept of Health, Education and Welfare (HEW)

Proposed test methods for determining performance of diagnostic X-Ray equipment.

(Application for copies should be addressed to Food & Drug Administration, Bureau of Radiological Health, Rockville, Md. 20852.)

3. REQUIREMENTS

3.1 Fire and casualty hazards.

3.1.1 The contractor shall submit proof, prior to delivery, that the items to be delivered under the contract conform to the applicable requirements of UL 187. Proof shall consist of one of the following:

A. UL label or UL listing of the product, provided the contractor agrees to have Underwriters' Laboratories, Inc. forward a copy of the UL report directly to the Contracting officer, upon the latter's request.

B. Laboratory report from a qualified independent laboratory. If an independent laboratory is used, it must be adequately equipped and competent to determine compliance with UL standards. The qualification of such laboratory shall be determined by the Government which reserves the right to require a facilities report and to conduct a facilities survey. Any laboratory report submitted shall include actual test methods, data taken, and results of tests. Detailed findings of examinations necessary to determine compliance shall also be included. The test report shall also have a section that describes the test sample in such a way that the report can be used to insure identity of production items with the test sample. This section shall be supplemented with construction information such as materials used, spacings between electrical components, types of bushings, photographs, wiring diagrams, sketches, and parts list, all appropriately identified. Compliance with this paragraph does not excuse compliance with all other requirements of the contract and specifications.

3.2 Materials. Materials used in the fabrication of the dental X-ray apparatus shall be of the highest quality throughout. All parts entering into the construction of the equipment shall be of sufficient strength to maintain operational alignment for the life of the equipment. Die-castings shall not be employed as load-bearing components of the equipment.

* 3.3 Design and construction. The dental X-ray apparatus shall be a wall mounted self-rectified type consisting of a master control assembly, a separate folding extension bracket for supporting and positioning the tubehead, a tubehead assembly, cones, mounting parts, and accessories. The control and extension arm assembly shall be designed for mounting on walls of various construction.

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The control assembly shall include a manual selector switch to permit operation of at least three remotely located tubeheads. The apparatus shall operate satisfactorily on a line voltage of 100 to 130 volts and shall be satisfactory for use on both 50 and 60 Hertz lines without the need for making any major modifications. The unit shall have a rating of at least 90 PKV at tube currents of 10 and 15 MA. All exposed welded parts shall be ground free of scale and protuberances to provide the best possible surface for paint or plating. All parts shall be suitably protected against corrosion by painting, plating, or equivalent means. All exposed parts subject to wear and handling shall have a durable wearing finish. All bolt connections shall be secured by lockwashers. Plastic parts such as knobs shall have molded metal reinforcements if set screws are utilized. Whenever applicable, screw threads shall be of the Unified Thread Series of ANSI Standard B1.1.

* 3.3.1 Radiation protection. The apparatus offered shall be in accordance with the requirements of NCRP Report No. 35. Any applicable sentence regarding x-ray apparatus using the words "shall", "should", or "advisable" shall be considered a requirement.

* 3.3.2 HEW Performance Standard. Units delivered after 1 August 1974 shall comply with latest requirements of the Department of Health, Education and Welfare (HEW) Standard for Diagnostic x-ray Systems; 21CFR, Part 278.

* 3.3.3 Control assembly. The components in the wall mounted master control assembly shall be housed in a cabinet of the "shelf" or similar design wherein front to back clearance directly below the mounted control unit extends to the wall on which the control is mounted. The master control assembly shall include the following components:

(a) Timer switch.
(b) Timer.
(c) MA and voltage selectors, and kilovolt (KV) indicating meter.
(d) Line compensator.
(e) MA stabilizer.
(f) Overload protection device.
(g) Ready light.
(h) Emission light.
(i) Main line switch.
(j) Inverse suppressor.
(k) Selector switch for minimum (min.) of three tubeheads.
(l) Junction box or terminal board.

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* 3.3.3.1 Timer switch. Timer switch for the master control shall be of the pushbutton, momentary-contact type, normally in the "off" position. The switch shall be enclosed in a durable protective case and shall be equipped with a spring-action-type coiled cable, expandable to a minimum length of 15 feet. There shall be no timer switch on the master control itself.

* 3.3.3.2 Timer. Timer shall be a transistorized or solid-state re-set type having solid-state contactors, with a minimum timing range of 1/60 to 5 seconds. A calibrated dial shall be mounted on the control and shall afford easy access for manually selecting the desired exposure time. There shall be not less than 22 mechanically indexed divisions in increments of approximate geometric progression on the dial. The time control circuit shall be compensated to account for the time lag required to heat the X-ray tube filament to preset emission temperature from a standby condition. From the lowest setting to 1/5 second, all exposures shall be correct to within complete impulses. An impulse shall be that portion of the sine wave starting within 32 and continuing to at least 148 degrees. Above 1/5 second to 1 second, the exposure shall be correct to within 1 impulse; except that on 50 Hz operation, at the 1 second setting, an accuracy of ± 2 pulses shall be acceptable. Above 1 second, the exposure shall be correct to within ± 3 percent. When a 3 pulse dial setting, based on 60 Hz current, is utilized on 50 Hz current, an elapsed time of 2 or 3 pulses shall be acceptable.

3.3.3.3 Voltage selector and KV meter. The master control unit shall be provided with an alternating current (AC) voltmeter calibrated directly in kilovolts. The KV meter dial shall be graduated and numbered for a minimum range of 65 to 90 peak kilovolts (PKV). PKV range may be greater, but shall, in all instances, fully reach 90 PKV at 15 milliamperes (MA). Intermediate graduations of not more than 2-1/2 kilovolts shall be indicated on the meter dial. The AC meter shall be accurate to within plus or minus 2 percent. A KV selector switch shall provide for the selection of kilovoltages in steps not greater than 2 PKV within a minimum range of 65 to 90 PKV, from taps on a transformer. As an alternate, stepless control of kilovoltage obtained by means of a variac and a prereading KV meter calibrated in not more than 2-1/2 KV increments may be provided.

3.3.3.3.1 Alternate type of voltage selection. As an alternate to the voltage selector and KV meter specified in 3.3.3.3, the master control unit may be supplied with a KV selector switch which shall provide for pushbutton type selection of kilovoltage in steps not greater than 5 PKV, within a minimum range of 65 to 90 PKV, from taps on a transformer. The pushbutton switch or bezel shall carry markings to indicate which kilovoltage setting has been selected. A kilovolt set meter, shall be furnished to indicate the standardized voltage to the tapped transformer for the two mandatory 10 MA and 15 MA settings

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and any other(s) included, as a calibration means for the PKV selector switch. When this alternate method is used, a milliammeter shall be provided in the control unit to indicate these settings. Milliampere adjustment shall be provided from a stepless control, calibrated to show the required 10 MA, 15 MA and any additional setting(s).

* 3.3.3.3.2 PKV accuracy. The PKV at 10 and 15 MA shall be within ± 4 percent of that shown on the KV meter dial or KV switch.

* 3.3.3.4 Line compensator. A line compensator shall be furnished and shall incorporate all wiring and circuitry necessary to match the incoming line, whether 50 or 60 Hz, to the autotransformer in establishing the volt per turn ratio required for efficient and safe operation of all circuits fed by the autotransformer. This matching shall be accomplished for at least three equal ranges of line voltage between 100 and 130 volts.

3.3.3.5 MA stabilizer.

* 3.3.3.5.1 Input voltage response. The MA output, both 10 and 15 MA, at 80 PKV, shall not vary more than ± 7 percent when an input voltage of 115 volts, 50 and 60 Hz is changed ± 8.7 percent.

* 3.3.3.5.2 PKV response. The MA output, both 10 and 15 MA, shall not vary more than ± 6.5 percent when a PKV of 77.5 is adjusted successively to 65 and 90 at an input voltage of 115 volts, 60 Hz; this requirement shall also apply for an input voltage of 100 and 130 volts, 50 and 60 Hz and 115 volts, 50 Hz.

* 3.3.3.6 Overload protection. Line overload protection in accordance with U.L. 187 shall be furnished. The line protector, and any other circuit protective devices furnished, shall be readily accessible to the operator for re-setting, or service.

* 3.3.3.7 Ready light. A pilot light shall be furnished on each master control, and each remote control when required, and at each tubehead location to indicate that the control unit and the selected tubehead is ready for use.

* 3.3.3.8 Emission light. A red emission light shall be provided on each master control, and at each tubehead location. This emission light shall be connected to illuminate only for the duration of an exposure. Where the alternate type of voltage selection (see 3.3.3.3.1) is provided, the milliammeter, which denotes exposure by the deflection of its needle, shall be an acceptable alternate to the red emission light on the master control unit.

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3.3.3.9 Main line switch. A main line switch which shall break both sides of the line shall be provided. The main line switch may be part of a combination with the overload protection device or may be separate. The action of this switch shall provide for quick-make and break-of the circuit with a minimum of arcing and burning of contact points. The design shall be to eliminate the possibility of "partial contact".

3.3.3.10 Inverse suppressor. Means to limit the inverse voltage on the tubehead transformer to provide protection to the tube and high-tension transformer against breakdown shall be furnished.

3.3.3.11 Tubehead selector switch. A switch shall be provided to permit selection of one of the tubehead assemblies desired for use. The switch shall indicate by numbered position which remote unit has been selected. As an alternate, a switch system including numbered indicator lights may be provided in which the illuminated light indicates the remote unit selected.

3.3.3.12 Junction box or terminal board. An electrical junction box or terminal board shall be provided for connection to the AC power source or line cord specified in 3.8, and for wiring to remote assemblies.

* 3.3.4 Tubehead-Tubearm assembly. The assembly shall consist of a wall mount, tubearm, and tubehead assembly.

3.3.4.1 Wall mount. The wall mount shall be designed and constructed to permit secure fastening of the arm assembly to a wall. The mount shall be a rigid plate or device provided with means to retain a vertical shaft on which the extension arm shall be swivel mounted. The mount shall also provide means for effecting permanent electrical connections from the control unit.

* 3.3.4.2 Tubearm assembly. The extension bracket shall be designed to support the tubehead adequately at any required position. The following movements shall be provided:

(a) 150 degrees minimum rotation (75 degrees minimum each side of center) around a vertical axis on the vertical post.

(b) With the arm assembly mounted so that the bracket is positioned in a plane perpendicular to both the wall and floor and with the yoke located in a plane parallel to the wall and perpendicular to the floor, the horizontal range of movement of the common centerline of the tubehead and cone assembly shall not be less than 25 inches. Movement shall be measured in a horizontal straight line from the center line of the vertical shaft in the wall mount.

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(c) The design of the extension arm shall permit a 17-inch-minimum vertical movement of the tubehead.

(d) The tubehead yoke shall rotate 360 degrees around its vertical axis. The tubehead shall rotate at least 310 degrees around a horizontal axis through the tubehead yoke. This rotation shall be so restricted that the cone shall not strike the tubehead yoke. However, if cones are made of shock-resistant material, they may touch the tubehead yoke at the extremes of rotation about the horizontal axis. All bracket hinge bearings, except the vertical bearing, shall be equipped with antifriction bearings of the roller or ball-bearing type. As an alternate, the overhead extension bracket may be equipped with roller bearings, oil-impregnated bronze bearings or other types for smooth easy action.

3.3.4.3 Tubehead assembly. The tubehead assembly shall be equipped with angulation scales, and shall consist of an X-ray tube, high-tension transformer, filament transformer, housing and two radiographic cones. The X-ray tube shall be designed for self-rectified operation. The tube shall be of the line focus type, and shall have a focal spot of 1.0 millimeters (mm.). The tube shall have adequate capacity to operate at 15 MA and 90 PKV. The tube shall be mounted by the anode stem or suitable brackets in such a manner that the X-ray beam shall be centered within the ray port of the tubehead. The window of the glass envelope shall be of uniform thickness with a filtration of the X-ray beam not to exceed 1/2 mm. of aluminum. The tube filament may be unheated or at a preheat temperature when the apparatus is turned on and at "standby". A means shall be provided to permit the tube filament to reach proper emitting temperature before an exposure can be made. The high-tension transformer shall be designed for self-rectified operation at a rated capacity of 15 MA over a minimum range of 65 to 90 PKV. The tubehead shall be capable of withstanding 125 percent of the rated peak kilovoltage at 15 MA, and the corresponding inverse KV, for a period of one minute. The filament transformer shall have sufficient capacity to energize the tube filament for operation at 10 MA and 15 MA. Adequate insulation shall be provided between the primary and secondary turns. The design shall be such that the filament of the tube cannot be damaged with the filament control set at its highest point and the filament energized continuously for 10 minutes with no high tension applied. The housing of the shockproof tubehead shall be of nonporous metal. This housing shall be designed to permit easy replacement of the X-ray tube. Provision shall be made for an X-ray port and cone support. The total filtration permanently in the useful beam shall be not less than 2.5 mm. aluminum equivalent. For oil insulated tubeheads, the housing shall accommodate a bellows to provide for the contraction and expansion of the installed oil with changing ambient and internal temperatures. Seams of the bellows shall be kept to a minimum. For gas insulated tubeheads, the housing shall be loaded with sulfur hexafluoride gas at a pressure of not less than 4.5 pounds per square inch. X-ray tube housing leakage radiation at a distance 1 meter from the target shall not exceed 100 milliroentgens in 1 hour when the tube is operated at 90 PKV and 15 MA.

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* 3.3.4.3.1 Cones. An 8-inch open-end radiographic cone and a 16-inch open-end radiographic long cone shall be provided. Each cone shall be fitted with a permanently fixed lead diaphragm to collimate the useful beam. The beam confining lead diaphragm shall be 3/32-inch in nominal thickness (2.38 mm. - 6 pounds per square foot). The diameter of the collimated beam at each cone tip shall not exceed 2-3/4 inches. The interior wall of each cone shall be coated with not less than 0.004 inch of lead to reduce scattered radiation.

* 3.3.4.3.2 Beam quality. The half-value layer (HVL) of the useful beam shall be not less than the values shown below:

Measured potential (PKV)	Half-value layer (mm of aluminum,) Type 1100 alloy
50	1.2
60	1.3
70	1.5
71	2.1
80	2.3
90	2.5
100	2.7

* 3.4 Operation calibration. Each unit shall pass the operation-calibration test specified in 4.4.7.

* 3.5 Duty cycle. The unit shall be capable of operating continuously for 4 hours at a minimum duty cycle of 4.4 percent, using 2-second intermittent exposures at input voltages of 100, 115, and 130 volts, 50/60 Hz. The MA shall not vary more than ± 1/2 MA throughout the 4 hour period of the test; the unit shall operate satisfactorily and shall show no signs of damage during or after the test.

* 3.6 Dielectric strength. Each unit shall pass the dielectric strength test specified in U.L. 187.

3.7 Mounting. The master control and tube-arm assembly shall be designed for wall mounting. All hardware necessary for mounting the master control unit and tube-arm assembly shall be furnished.

3.8 Line cord. A three-wire line cord, not less than 12 feet in length, shall be provided along with a polarized plug and receptacle.

* 3.9 Finish. All exposed metal parts to be painted shall be covered with a bonderizing pretreatment coat followed by two coats of baked-on enamel. All metal parts not painted shall be chromium plated. Nonmetallic parts shall be black or cream white in color (with the exception of the cones, which may be supplied in

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the color normally furnished by the manufacturer) and shall be finished in accordance with the best commercial practice. As an alternate, the exposed metal parts shall be covered by at least 2 coats of baked-on primer followed by sprayed-on vinyl or hot lacquer. The lacquer or vinyl shall be at least 0.003 inch thick, and shall be hard and durable.

3.9.1 Color. The paint finish shall be cream white, dental, color 13531 (see 6.4) of Federal Standard No. 595.

3.10 Identification marking. The master control of each apparatus shall be permanently stamped with, or shall have a metal nameplate(s) permanently affixed and marked with, the following information:

National Stock number
Serial number.
Manufacturer's name or registered trademark.
Part or model number.
Contract or purchase order number
Electrical rating (amperes, volts).
U.S.

* 3.10.1 Additional marking. The master control shall contain permanent marking to show the procedure necessary to convert from 60 Hz to 50 Hz operation; and a warning label visible to the operator advising of possible danger to patient and operator unless safe exposure factors and operating instructions are observed.

* 3.11 Service Data. The contractor shall furnish, with each unit, two copies each of a technical manual which contains complete instructions for installation, operation, and maintenance and lists of component parts. The manual shall comply with the requirements of MIL-M-7298. As soon as practicable after award of the contract or purchase order, the contractor shall furnish to the Contracting Officer, or the cognizant Government inspector, two preliminary copies of his proposed technical manual, for review and comment by the Directorate of Medical Materiel, Defense Personnel Support Center. The preliminary manuals shall be submitted sufficiently early to permit adequate review by the Directorate of Medical Materiel, Defense Personnel Support Center (based on a maximum period of 30 days for such review) and to allow incorporation by the contractor of any required revisions and corrections in the final manual without delaying delivery under the contract or purchase order. In addition, the Contractor shall furnish 15 copies of the approved technical manual to the Contracting Officer for distribution. Should the supplier have furnished acceptable manuals for the identical item within two years preceding the date of contract and propose to furnish manuals identical with those previously accepted, then the requirements for review of drafts and submission of 15 additional copies may be waived upon request to the Contracting Officer. Waiver of submission of 15 additional copies shall be at the option of the Government.

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3.11.1 Additional data. Service data shall also include a statement of caution, prominently displayed, to the effect that the apparatus shall be used without a collimating cone only for extra-oral radiography where a larger than normal usable beam is required, and that a cone shall be replaced immediately upon completion of extra-oral radiography. The statement shall also include a warning against the removal of the aluminum filter within the cone opening of the tube housing for any type of radiography.

3.12 Interchangeability. Any component or part incorporated in any unit of this X-ray apparatus shall be completely interchangeable, without modification, with the corresponding component or part in any other unit in the entire production of this apparatus under a single contract or order. Any deviation from this requirement proposed by the contractor shall be submitted to the contracting officer for decision prior to its incorporation in the apparatus. Accordingly, any tubehead produced under a given contract or order shall operate satisfactorily, without modification, with any control manufactured under the same contract or order. Similarly, any control manufactured under a given contract or order shall operate satisfactorily with any tubehead manufactured under the same contract or order.

3.13 Tools and equipment. The construction of the X-ray apparatus shall be such that no special tools and test equipment peculiar to the apparatus shall be necessary for satisfactory installation and calibration.

3.14 Workmanship. Workmanship, finish, and safety shall be first class throughout. The dental units shall be well made without any operating or other faults and the workmanship shall be free from defects which detract from their appearance or which may impair their function or serviceability. All exposed corners and edges shall be broken, rounded, or otherwise treated to eliminate sharp edges.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual, and auditory, investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the raw materials may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The Certificate shall include the name of the contractor, the contract number, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

* 4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with Table I and MIL-STD-105.

Table I

For visual examination	Inspection Level	AQL (defects per hundred units)
Major defects	II	1.0
Minor defects	II	4.0
Dimensional examination	S-3	2.5

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and Table II. Acceptance number shall be zero for all tests.

Table II

Requirement	Test method	Inspection Level
Timer (3.3.3.2)	4.4.1	S-2*
KV accuracy (3.3.3.3.2)	4.4.2	**
MA stabilization (3.3.3.5.1) to input voltage	4.4.3.1	S-1*
MA stabilization (3.3.3.5.2) to KVP	4.4.3.2	S-1*

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Table II (Cont'd)

Requirement	Test method	Inspection Level
Focal spot size (3.3.4.3)	4.4.4	100 percent
Tubehead leakage (3.3.4.3)	4.4.5	one from each lot
Beam quality (3.3.4.3.1)	4.4.6	**
Operation-calibration (3.4)	4.4.7	100 percent
Duty cycle (3.5)	4.4.8	**
Dielectric strength (3.6)	4.4.9	100 percent

* 50 Hz test results shall be required for one unit from each lot.

** Should the supplier have previously furnished an acceptable test report and propose to furnish an item identical with that previously accepted, then the contractor need not perform test 4.4.2, 4.4.6 and 4.4.8 to show compliance for any future contract.

* 4.3 Examination. The X-ray apparatus shall be examined to determine compliance with all requirements contained in this specification.

4.3.1 Classification of defects. Examination shall be conducted in accordance with the following classification of defects.

Categories and defects*
Major
101 Apparatus not complete with all required parts and subassemblies.
102 Not style and design specified.
103 Mating parts and subassemblies not properly fitted and aligned.
104 Welds and soldered joints and electrical connections not sound, smooth, and free of flux, spatter, and discoloration.
105 Exposed edges not free of burrs and sharpness.
106 Parts not free of cracks, tears, ruptures, and deformed areas.
107 Identification marking not correct, complete, legible.
108 Not free of burrs, sharp edges.
Minor
201 Identification marking not present, complete, legible, correct.
202 Finish not free of scratches, mars, chips.
203 Not clean, free of grease,oil dirt, other foreign matter.
204 Tubearm does not operate properly.
205 Color not as specified.
206 Service data not complete, missing.
207 Fasteners or rivets not tight, properly aligned.

*Examination shall not be restricted to the classified possible defects listed above.

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* 4.4 Tests. Unless otherwise specified herein, the X-ray unit shall be operated in accordance with the manufacturer's directions at an ambient temperature of 70° to 80° F. All electrical components of the unit shall be operable during performance tests. MA settings shall be within ± 0.1 MA of the required setting. Input power settings shall be maintained within ± 1 percent throughout the specified tests. Test meters and equipment shall be in accordance with the requirements of GG-X-635.

* 4.4.1 Timer. Timer accuracy shall be determined in accordance with the method in GG-X-635, at settings of 1/60, 1/15, 1/6, and 1/2 second, and 1, 3 and 5 seconds, at an input voltage of 115 volts, 60 Hz, and the corresponding 50 Hz settings at an input voltage of 115 volts, 50 Hz.

* 4.4.2 KV accuracy. The accuracy of the KVP output shall be determined at approximate settings of 70, 80 and 90 KVP, at tube currents of 10 and 15 MA, with an input voltage of 115 volts, 50 and 60 Hz. Test procedure shall be in accordance with the method in the publication "Radiology" Vol. 74, No. 2, pages 255 to 264, titled "Influence of Cable Length on Dose Rate and Half-Value Layer in Diagnostic X-ray Procedures".

4.4.3 MA stabilization.

* 4.4.3.1 Input voltage variations. Prior to testing, the input power leads shall be connected to the unit, in accordance with the manufacturer's directions for a 115 volt, 60 Hz power input, and shall not be again moved as the input voltage is adjusted to higher and lower values. The MA shall be calibrated to 10 and 15 MA ± 0.1 MA, under load, at a KVP of 80 at the 115 volt, 60 Hz input. The KVP shall be adjusted to 80 KVP, under load, prior to each measurement of MA. Exposures shall be one second in duration, and at least one exposure shall be made each minute. The MA output, both 10 and 15 MA, shall be measured at 115 volts, 105 volts and 125 volts 60 Hz. as specified above, to determine conformance to 3.3.3.5.1. The test shall then be repeated as specified above for input voltages of 115, 105, and 125 volts, 50 Hz.

* 4.4.3.2 KV variations. An input power of 115 volts, ± 1 percent, 60 Hz shall be connected to the unit in accordance with the manufacturer's directions. Prior to testing, the unit shall be calibrated under load to 10 and 15 MA ± 0.1 MA at 77.5 KVP. The MA shall not be again calibrated at any time during the test. Exposures shall be 1 second in duration and at least one exposure shall be made each minute. The MA output shall then be measured successively at 65, 77.5 and 90 KVP, under load, to determine conformance to 3.3.3.5.2. This test shall be repeated for an input voltage of 100 and 130 volts, ± 1 percent, 50 and 60 Hz; and 115 volts, ± 1 percent 50 Hz.

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* 4.4.4 Focal spot size. Focal spot shall be measured as specified in FED-STD-83. Pin hole sizes shall conform to the requirements of IEC Publication 336.

* 4.4.5 Tubehead leakage. Radiation leakage shall be determined in accordance with the HEW Proposed methods for determining compliance with the Diagnostic X-ray equipment performance standard, dated 6 Mar, 1972.

* 4.4.6 Beam Quality. Beam quality shall be determined for the applicable PKV values shown in 3.3.4.3.1 in accordance with the HEW proposed test methods for performance of X-ray equipment dated 6 Mar 1972.

* 4.4.7 Operation-calibration. The dental X-ray unit shall be operated at an input voltage of 115 volts, 60 Hz to determine that all controls, components, and meters are functioning properly, and that the unit is properly calibrated.

* 4.4.8 Duty cycle. The unit shall be operated continuously for 4 hours at 90 PKV, under load, and 15 MA, at the duty cycle specified in 3.5, or if a higher duty cycle is recommended by the manufacturer, at the higher duty cycle. The 4 hour duty cycle test shall be performed at input voltages of 100 volts, 50 and 60 Hz; with the line compensator adjusted according to the manufacturer's directions. The MA and KVP shall be recorded every 20 minutes throughout the test.

* 4.4.9 Dielectric strength. The Dielectric strength test shall be performed in accordance with the test method in U.L. 187.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be Level A or C as specified (see 6.2).

5.1.1 Unit of issue. One X-ray apparatus, as specified, constitutes one unit of issue.

5.1.2 Level A.

5.1.2.1 Unit packaging. Each unit shall be disassembled as far as it is practicable. The component parts and accessories shall be wrapped in kraft paper, cushioned with flexible corrugated fiberboard or other suitable materials. All wrapping and cushioning materials shall be secured with cotton cord or tape, or by sealing with kraft gummed paper tape. The component parts, accessories, or disassembled parts, except X-ray tube, shall be further packaged in one or more fiberboard boxes of appropriate size and design and constructed in accordance with PPP-B-636, type CF, class domestic.

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5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be Level A or C as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Exterior container. Each X-ray apparatus and each X-ray tube shall be packed separately in an exterior container designed for type 3 load, and constructed in accordance with the requirements of PPP-B-621, class 2 or PPP-B-601, overseas type. Closure and strapping shall comply with appendix of applicable box specification. Each box shall be lined with a waterproof barrier fabricated of material conforming to the requirements of PPP-B-1055, class L-2(b) or M-1, and sealed with a suitable waterproof adhesive. The X-ray apparatus and the X-ray tube shall be adequately secured within its own exterior container, by means of bolts, wood braces, blocks, or other suitable devices, to insure against shifting and breakage or marring of surfaces. No excelsior, hay or straw shall be used for cushioning components within the container.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Interior packages. Each container shall be marked to identify contents.

5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The dental X-ray unit is intended for use in all cephalometric and intraoral radiography.

6.2 Ordering data. Procurement documents should specify the following:

(a) Number and date of this specification.
(b) Selection of applicable levels of packing (see 5.1 and 5.2).

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6.3 This specification covers the following item appearing in the Federal Supply Catalog:

National Stock Number	Item Identification
6525-00-975-2509	X-RAY APPARATUS, DENTAL WALL MOUNTED

6.4 Availability of color chips. The individual 3 by 5 inch chips of the color referenced in 3.9.1 are available from General Services Administration, Business Service Center, Region 3, Seventh and D Streets, S.W. Washington, D. C., 20225.

6.5 This specification does not cover all types, classes, grades, or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the types which are normally procured to meet requirements.

6.6 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Army - MD
Navy - MS
Air Force - 03

Review activities:

Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DSA-DM

Project No. 6525-1117

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